ESTIMATIONS & RATIONALE DOCUMENT

MBA 666

Green Hydrogen Production Plant

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Document Purpose:

To detail the complete Capital Expenditure (CAPEX), forecast the annual Operating Expenditure (OPEX), and model the project's financial viability through Levelized Cost of Hydrogen (LCOH), Net Present Value (NPV), Internal Rate of Return (IRR), and Payback Period (PBP)

1. Capital Expenditure(CAPEX)

The CAPEX is the total one-time cost to build the plant. This budget is derived directly from the Project Scope Document (V1.1) and includes all necessary hardware, integration, and contingency.

Main Component	Sub-Component	Cost (₹ crore)
Solar PV Plant (2.5 MWp) **(C-1)**	Panels, Inverters, Structures, Land	₹ 8.75
Electrolyzer System (1 MW) **(P-1 & P-2)**	1 MW Alkaline Stack, Rectifier, BoP	₹ 5.50
Battery Storage (BESS, 1 MWh) **(C-2)**	BESS Unit, Power Conversion System	₹ 2.00

Storage & Safety Systems **(P-3)**	Compressor (350 bar)	₹ 1.20
(P-3 & S-1)	Cascade Storage Vessels (1.5 ton)	₹ 0.80
(S-2) (NFPA 2, ESD systems)	Safety/Fire/Gas Systems	₹ 0.50
	Sub-Total	**₹ 2.50**
Civil & Utility Works **(U-1)**	Water Purification (RO/MBDI)	₹ 0.40
(S-4)	Civil Works (Foundations, Control Room)	₹ 0.70
(C-3)	Grid Integration & SCADA	₹ 0.40
	Sub-Total	**₹ 1.50**
EPC & Contingency **(10% project buffer)**	EPCM, Project Management, Training	₹ 2.03
Total Estimated CAPEX		**₹ 22.28 crore**

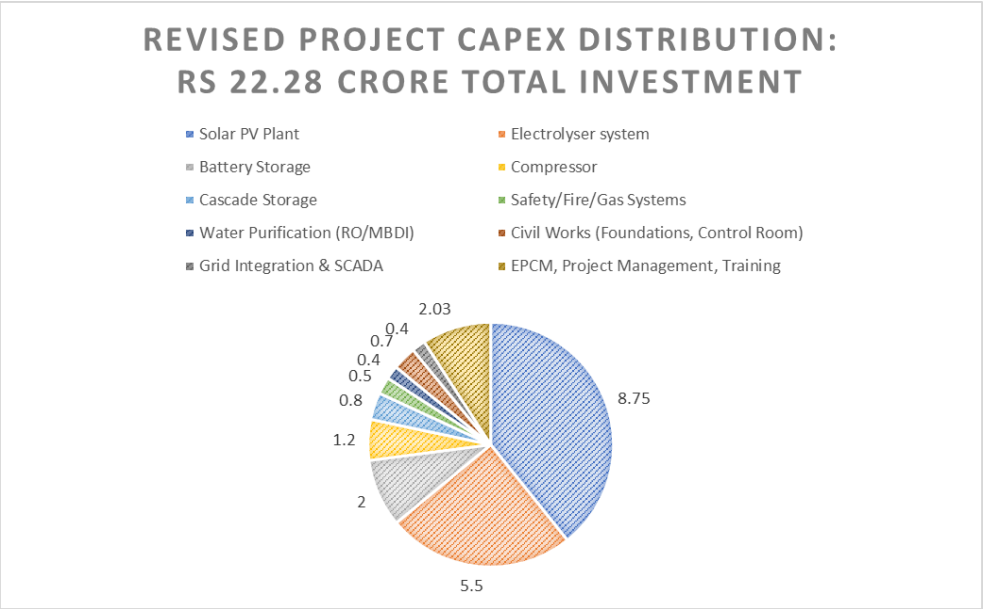

Fig: Revised project CAPEX distribution

2. Operating Expenditure(OPEX)

This section itemizes the annual costs (Total: ₹2.324 crore) required to run the plant, based on the requirements.

Energy Cost(Total : ₹1.752 Cr)

Item	Rationale / Calculation	Annual Cost (₹)
Grid Electricity	(24 MWh Req. – 12 MWh Solar) × 365 days × ₹4.0/kWh	₹ 1,75,20,000

Utility Cost(Total : ₹0.015 Cr)

Item	Rationale / Calculation	Annual Cost (₹)
Raw Water	(10 m^3/day) × (365 days) × (₹40/m^3)	₹ 1,46,000

Maintenance & Labor Costs (Total : ₹0.557 Cr)

Item	Rationale	Annual Cost (₹)
Maintenance Labor	2 full-time technicians/engineers.	₹ 20,00,000
Spares & Consumables	Electrolyzer parts, RO filters, resin, lubricants.	₹ 30,00,000
Insurance & Admin	Site insurance, admin overhead.	₹ 5,70,000

3. Levelized Cost of Hydrogen (LCOH)

LCOH is the "break even" price per kg over the project's 20-year life.

Parameter	**Calculation**	**Value (₹)**
A. Annualized CAPEX	PMT(rate = 10%, nper = 20, PV = –22.28 Cr)	₹ 2,61,70,000
B. Total Annual OPEX	Energy + Utility + Maintenance	₹ 2,32,40,000
C. Total Annualized Cost	A + B	₹ 4,94,10,000
D. Annual H_2 Production	480 kg/day × 365 days	1,75,200 kg

LCOH(C/D)	C / D = ₹ 4,94,10,000 / 1,75,200 kg	**₹ 282.02 / kg**

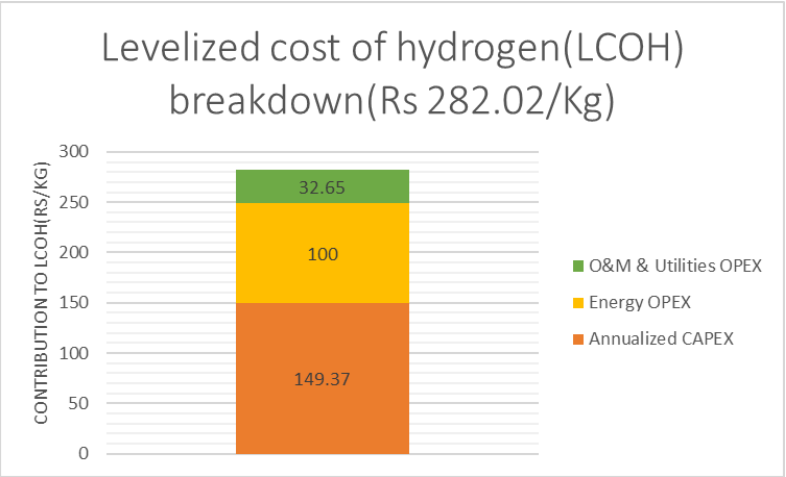

Fig: Levelized cost of Hydrogen breakdown

4. Financial Viability Analysis (NPV, IRR, PBP)

This analysis determines if the project is a profitable investment, assuming a hydrogen selling price of ₹350 per kg.

I. Annual Cash Flow

Annual Cash Flow	Calculation	Value (₹ crore)
Annual Revenue	175,200 kg/year × ₹ 350/kg	₹ 6.132
Annual OPEX	From Section 2	₹ 2.324
Annual Profit	Revenue – OPEX	₹ 3.808

II. Cumulative Discounted Cash Flow

Year	Cash Flow	discount factor	DCF	CDCF
0	-222800000	1	-222800000	-222800000
1	38080000	0.909090909	34618181.8	-188181818
2	38080000	0.826446281	31471074.4	-156710744
3	38080000	0.751314801	28610067.6	-128100676
4	38080000	0.683013455	26009152.4	-102091524
5	38080000	0.620921323	23644684	-78446840
6	38080000	0.56447393	21495167.3	-56951673
7	38080000	0.513158118	19541061.1	-37410611
8	38080000	0.46650738	17764601	-19646010
9	38080000	0.424097618	16149637.3	-3496373.1
10	38080000	0.385543289	14681488.5	11185115.4
11	38080000	0.350493899	13346807.7	24531923.1
12	38080000	0.318630818	12133461.5	36665384.6
13	38080000	0.28966438	11030419.6	47695804.2
14	38080000	0.263331254	10027654.2	57723458.4
15	38080000	0.239392049	9116049.24	66839507.6
16	38080000	0.217629136	8287317.49	75126825.1
17	38080000	0.197844669	7533924.99	82660750.1
18	38080000	0.17985879	6849022.72	89509772.8
19	38080000	0.163507991	6226384.29	95736157.1
20	38080000	0.148643628	5660349.36	101396506

Fig: Cumulative discounted cash flow values over 20 years

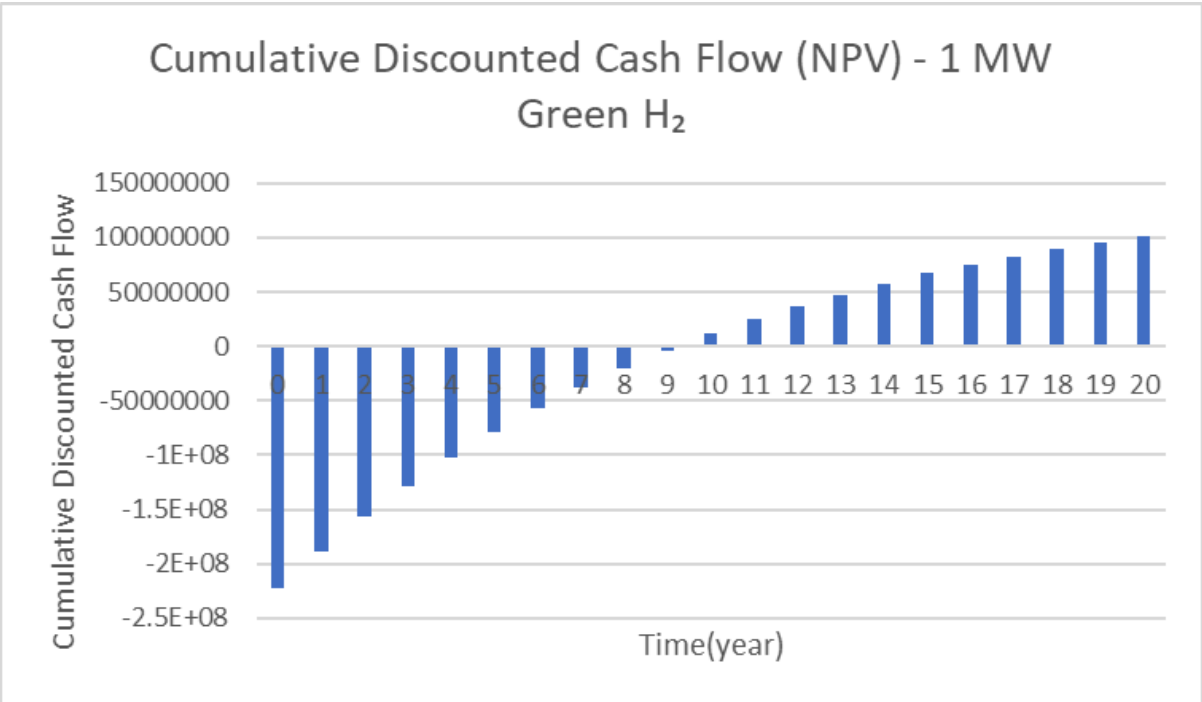

Fig: Bar graph depicting CDCF vs Time

III. Viability Metrics

The financial analysis, including NPV and IRR, is modeled over a **20-year project life**, which is the standard operational horizon for utility-scale solar PV and electrolyzer plants, assuming mid-life stack replacement. The **$10% Discount Rate** (or Cost of Capital) is a conservative estimate based on the average Weighted Average Cost of Capital (WACC) for similar, medium-risk industrial infrastructure projects in India's current financial climate.

1. Payback Period (PBP)

Formula: Total Initial Investment (CAPEX) ÷ Annual Profit (Cash Flow)
Calculation:₹22.28 crore ÷ ₹3.808 crore/year
Result: 5.85 years

2. Net Present Value (NPV)

$$\text{NPV} = \left[\sum_{t=1}^{n} \frac{\text{CF}_t}{(1+r)^t}\right] - C_0$$

(Where CF_t = Cash Flow in year t,r = discount rate,n = life, C_0 = CAPEX)

Step 1: Calculate the Present Value (PV) of all future profits:
PV = (₹3.808 Cr) × [(1 - (1 + 0.10)^-20) / 0.10]
PV = (₹3.808 Cr) × [(1 - (1.10)^-20) / 0.10]
PV = (₹3.808 Cr) × [(1 - 0.14864) / 0.10]
PV = (₹3.808 Cr) × [0.85136 / 0.10]
PV = (₹3.808 Cr) × 8.5136

PV of Future Profits = ₹32.42 crore

Step 2:Calculate the NPV:
NPV = (PV of Future Profits) - (Initial Investment)
NPV = ₹32.42 crore - ₹22.28 crore

Result: + ₹10.14 crore

2. Internal Rate of Return (IRR)

The project's inherent percentage return. The project is viable if IRR > Discount Rate (10%).

$$₹0 = \left[\sum_{t=1}^{n} \frac{\text{CF}_t}{(1+r)^t}\right] - C_0$$

22.28 Cr (CAPEX) = (3.808 Cr) × [(1 - (1 + r)^-20) / r]

Code:

```
% Given values
CAPEX = 22.28;        % in crore
CF = 3.808;           % annual cash flow in crore
n = 20;               % project life in years
% NPV function in terms of r
npv_fun = @(r) CF * (1 - (1 + r)^(-n)) / r - CAPEX;
% Initial guess
r_guess = 0.1;
% Solve IRR using fzero
IRR = fzero(npv_fun, r_guess);
% Convert to percentage
IRR_percent = IRR * 100;
disp(['IRR = ', num2str(IRR_percent), '%']);
```

Result:

```
>> irr_function
IRR = 16.2504%
```